

In Problems 49 and 50, use the substitution  $t = -x$  to solve the given initial-value problem on the interval  $(-\infty, 0)$ .

49.  $4x^2y'' + y = 0$ ,  $y(-1) = 2$ ,  $y'(-1) = 4$

50.  $x^2y'' - 4xy' + 6y = 0$ ,  $y(-2) = 8$ ,  $y'(-2) = 0$

### Contributed Problem

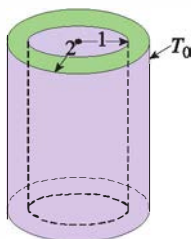
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51. **Temperature of a Fluid** A very long cylindrical shell is formed by two concentric circular cylinders of different radii. A chemically reactive fluid fills the space between the concentric cylinders as shown in green in **FIGURE 3.6.2**. The inner cylinder has a radius of 1 and is thermally insulated, while the outer cylinder has a radius of 2 and is maintained at a constant temperature  $T_0$ . The rate of heat generation in the fluid due to the chemical reactions is proportional to  $T/r^2$ , where  $T(r)$  is the temperature of the fluid within the space bounded between the cylinders defined by  $1 < r < 2$ . Under these conditions the temperature of the fluid is defined by the following boundary-value problem:

$$\frac{1}{r} \frac{d}{dr} \left( r \frac{dT}{dr} \right) = \frac{T}{r^2}, \quad 1 < r < 2,$$

$$\left. \frac{dT}{dr} \right|_{r=1} = 0, \quad T(2) = T_0.$$

- (a) Find the temperature distribution  $T(r)$  within the fluid.  
(b) Find the minimum and maximum values of  $T(r)$  on the interval defined by  $1 \leq r \leq 2$ . Why do these values make intuitive sense?



**FIGURE 3.6.2** Cylindrical shell in Problem 51

### Discussion Problems

52. Find a Cauchy–Euler differential equation of lowest order with real coefficients if it is known that 2 and  $1 - i$  are two roots of its auxiliary equation.  
53. The initial conditions  $y(0) = y_0$ ,  $y'(0) = y_1$ , apply to each of the following differential equations:

$$x^2y'' = 0,$$

$$x^2y'' - 2xy' + 2y = 0,$$

$$x^2y'' - 4xy' + 6y = 0.$$

For what values of  $y_0$  and  $y_1$  does each initial-value problem have a solution?

54. What are the  $x$ -intercepts of the solution curve shown in Figure 3.6.1? How many  $x$ -intercepts are there in the interval defined by  $0 < x < \frac{1}{2}$ ?

### Computer Lab Assignments

In Problems 55–58, solve the given differential equation by using a CAS to find the (approximate) roots of the auxiliary equation.

55.  $2x^3y''' - 10.98x^2y'' + 8.5xy' + 1.3y = 0$   
56.  $x^3y''' + 4x^2y'' + 5xy' - 9y = 0$   
57.  $x^4y^{(4)} + 6x^3y''' + 3x^2y'' - 3xy' + 4y = 0$   
58.  $x^4y^{(4)} - 6x^3y''' + 33x^2y'' - 105xy' + 169y = 0$

In Problems 59 and 60, use a CAS as an aid in computing roots of the auxiliary equation, the determinants given in (10) of Section 3.5, and integrations.

59.  $x^3y''' - x^2y'' - 2xy' + 6y = x^2$   
60.  $x^3y''' - 2x^2y'' - 8xy' + 12y = x^{-4}$

## 3.7 Nonlinear Equations

**Introduction** The difficulties that surround higher-order *nonlinear* DEs and the few methods that yield analytic solutions are examined next.

**Some Differences** There are several significant differences between linear and nonlinear differential equations. We saw in Section 3.1 that homogeneous linear equations of order two or

higher have the property that a linear combination of solutions is also a solution (Theorem 3.1.2). Nonlinear equations do not possess this property of superposability. For example, on the interval  $(-\infty, \infty)$ ,  $y_1 = e^x$ ,  $y_2 = e^{-x}$ ,  $y_3 = \cos x$ , and  $y_4 = \sin x$  are four linearly independent solutions of the nonlinear second-order differential equation  $(y'')^2 - y^2 = 0$ . But linear combinations such as  $y = c_1 e^x + c_3 \cos x$ ,  $y = c_2 e^{-x} + c_4 \sin x$ ,  $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x$  are not solutions of the equation for arbitrary nonzero constants  $c_i$ . See Problem 1 in Exercises 3.7.

In Chapter 2 we saw that we could solve a few nonlinear first-order differential equations by recognizing them as separable, exact, homogeneous, or perhaps Bernoulli equations. Even though the solutions of these equations were in the form of a one-parameter family, this family did not, as a rule, represent the general solution of the differential equation. On the other hand, by paying attention to certain continuity conditions, we obtained general solutions of linear first-order equations. Stated another way, nonlinear first-order differential equations can possess singular solutions whereas linear equations cannot. But the major difference between linear and nonlinear equations of order two or higher lies in the realm of solvability. Given a linear equation there is a chance that we can find some form of a solution that we can look at, an explicit solution or perhaps a solution in the form of an infinite series. On the other hand, nonlinear higher-order differential equations virtually defy solution. This does not mean that a nonlinear higher-order differential equation has no solution but rather that there are no analytical methods whereby either an explicit or implicit solution can be found.

Although this sounds disheartening, there are still things that can be done; we can always analyze a nonlinear DE qualitatively and numerically.

Let us make it clear at the outset that nonlinear higher-order differential equations are important—dare we say even more important than linear equations?—because as we fine-tune the mathematical model of, say, a physical system, we also increase the likelihood that this higher-resolution model will be nonlinear.

We begin by illustrating an analytical method that occasionally enables us to find explicit/implicit solutions of special kinds of nonlinear second-order differential equations.

**Reduction of Order** Nonlinear second-order differential equations  $F(x, y', y'') = 0$ , where the dependent variable  $y$  is missing, and  $F(y, y', y'') = 0$ , where the independent variable  $x$  is missing, can sometimes be solved using first-order methods. Each equation can be reduced to a first-order equation by means of the substitution  $u = y'$ .

The next example illustrates the substitution technique for an equation of the form  $F(x, y', y'') = 0$ . If  $u = y'$ , then the differential equation becomes  $F(x, u, u') = 0$ . If we can solve this last equation for  $u$ , we can find  $y$  by integration. Note that since we are solving a second-order equation, its solution will contain two arbitrary constants.

### EXAMPLE 1 Dependent Variable $y$ Is Missing

Solve  $y'' = 2x(y')^2$ .

**SOLUTION** If we let  $u = y'$ , then  $du/dx = y''$ . After substituting, the second-order equation reduces to a first-order equation with separable variables; the independent variable is  $x$  and the dependent variable is  $u$ :

$$\frac{du}{dx} = 2xu^2 \quad \text{or} \quad \frac{du}{u^2} = 2x \, dx$$

$$\int u^{-2} du = \int 2x \, dx$$

$$-u^{-1} = x^2 + c_1^2.$$

The constant of integration is written as  $c_1^2$  for convenience. The reason should be obvious in the next few steps. Since  $u^{-1} = 1/y'$ , it follows that

$$\frac{dy}{dx} = -\frac{1}{x^2 + c_1^2}$$

and so 
$$y = -\int \frac{dx}{x^2 + c_1^2} \quad \text{or} \quad y = -\frac{1}{c_1} \tan^{-1} \frac{x}{c_1} + c_2.$$

Next we show how to solve an equation that has the form  $F(y, y', y'') = 0$ . Once more we let  $u = y'$ , but since the independent variable  $x$  is missing, we use this substitution to transform the differential equation into one in which the independent variable is  $y$  and the dependent variable is  $u$ . To this end we use the Chain Rule to compute the second derivative of  $y$ :

$$y'' = \frac{du}{dx} = \frac{du}{dy} \frac{dy}{dx} = u \frac{du}{dy}.$$

In this case the first-order equation that we must now solve is  $F(y, u, u \, du/dy) = 0$ .

### EXAMPLE 2 Independent Variable $x$ Is Missing

Solve  $yy'' = (y')^2$ .

**SOLUTION** With the aid of  $u = y'$ , the Chain Rule shown above, and separation of variables, the given differential equation becomes

$$y \left( u \frac{du}{dy} \right) = u^2 \quad \text{or} \quad \frac{du}{u} = \frac{dy}{y}.$$

Integrating the last equation then yields  $\ln |u| = \ln |y| + c_1$ , which, in turn, gives  $u = c_2 y$ , where the constant  $\pm e^{c_1}$  has been relabeled as  $c_2$ . We now resubstitute  $u = dy/dx$ , separate variables once again, integrate, and relabel constants a second time:

$$\int \frac{dy}{y} = c_2 \int dx \quad \text{or} \quad \ln |y| = c_2 x + c_3 \quad \text{or} \quad y = c_4 e^{c_2 x}. \quad \equiv$$

**Use of Taylor Series** In some instances a solution of a nonlinear initial-value problem, in which the initial conditions are specified at  $x_0$ , can be approximated by a Taylor series centered at  $x_0$ .

### EXAMPLE 3 Taylor Series Solution of an IVP

Let us assume that a solution of the initial-value problem

$$y'' = x + y - y^2, \quad y(0) = -1, \quad y'(0) = 1 \quad (1)$$

exists. If we further assume that the solution  $y(x)$  of the problem is analytic at 0, then  $y(x)$  possesses a Taylor series expansion centered at 0:

$$y(x) = y(0) + \frac{y'(0)}{1!}x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!}x^3 + \frac{y^{(4)}(0)}{4!}x^4 + \frac{y^{(5)}(0)}{5!}x^5 + \cdots \quad (2)$$

Note that the value of the first and second terms in the series (2) are known since those values are the specified initial conditions  $y(0) = -1$ ,  $y'(0) = 1$ . Moreover, the differential equation itself defines the value of the second derivative at 0:  $y''(0) = 0 + y(0) - y(0)^2 = 0 + (-1) - (-1)^2 = -2$ . We can then find expressions for the higher derivatives  $y'''$ ,  $y^{(4)}$ , ..., by calculating the successive derivatives of the differential equation:

$$y'''(x) = \frac{d}{dx} (x + y - y^2) = 1 + y' - 2yy' \quad (3)$$

$$y^{(4)}(x) = \frac{d}{dx} (1 + y' - 2yy') = y'' - 2yy'' - 2(y')^2 \quad (4)$$

$$y^{(5)}(x) = \frac{d}{dx} (y'' - 2yy'' - 2(y')^2) = y''' - 2yy''' - 6y'y'' \quad (5)$$

and so on. Now using  $y(0) = -1$  and  $y'(0) = 1$  we find from (3) that  $y'''(0) = 4$ . From the values  $y(0) = -1$ ,  $y'(0) = 1$ , and  $y''(0) = -2$ , we find  $y^{(4)}(0) = -8$  from (4). With the additional information that  $y'''(0) = 4$ , we then see from (5) that  $y^{(5)}(0) = 24$ . Hence from (2), the first six terms of a series solution of the initial-value problem (1) are

$$y(x) = -1 + x - x^2 + \frac{2}{3}x^3 - \frac{1}{3}x^4 + \frac{1}{5}x^5 + \cdots \quad \equiv$$

**Use of a Numerical Solver** Numerical methods, such as Euler's method or a Runge-Kutta method, are developed solely for first-order differential equations and then are extended to systems of first-order equations. In order to analyze an  $n$ th-order initial-value problem numerically, we express the  $n$ th-order ODE as a system of  $n$  first-order equations. In brief, here is how it is done for a second-order initial-value problem: First, solve for  $y''$ ; that is, put the DE into normal form  $y'' = f(x, y, y')$ , and then let  $y' = u$ . For example, if we substitute  $y' = u$  in

$$\frac{d^2y}{dx^2} = f(x, y, y'), \quad y(x_0) = y_0, \quad y'(x_0) = u_0, \quad (6)$$

then  $y'' = u'$  and  $y'(x_0) = u(x_0)$  so that the initial-value problem (6) becomes

$$\begin{aligned} \text{Solve: } & \begin{cases} y' = u \\ u' = f(x, y, u) \end{cases} \\ \text{Subject to: } & y(x_0) = y_0, \quad u(x_0) = u_0. \end{aligned}$$

However, it should be noted that a commercial numerical solver *may not* require\* that you supply the system.

#### EXAMPLE 4 Graphical Analysis of Example 3

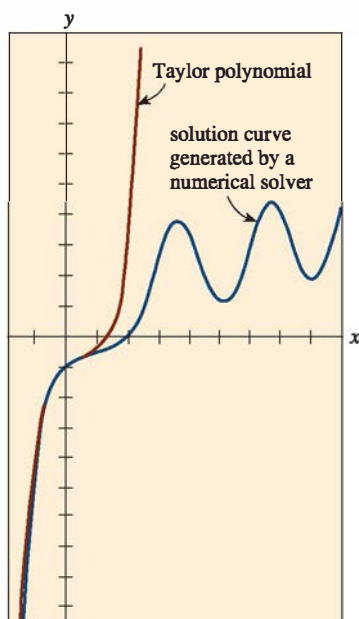
Following the foregoing procedure, the second-order initial-value problem in Example 3 is equivalent to

$$\begin{aligned} \frac{dy}{dx} &= u \\ \frac{du}{dx} &= x + y - y^2 \end{aligned}$$

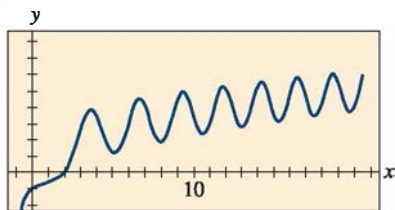
with initial conditions  $y(0) = -1$ ,  $u(0) = 1$ . With the aid of a numerical solver we get the solution curve shown in blue in **FIGURE 3.7.1**. For comparison, the curve shown in red is the graph of the fifth-degree Taylor polynomial  $T_5(x) = -1 + x - x^2 + \frac{2}{3}x^3 - \frac{1}{3}x^4 + \frac{1}{5}x^5$ . Although we do not know the interval of convergence of the Taylor series obtained in Example 3, the closeness of the two curves in a neighborhood of the origin suggests that the power series may converge on the interval  $(-1, 1)$ . ≡

**Qualitative Questions** The blue graph in Figure 3.7.1 raises some questions of a qualitative nature: Is the solution of the original initial-value problem oscillatory as  $x \rightarrow \infty$ ? The graph generated by a numerical solver on the larger interval shown in **FIGURE 3.7.2** would seem to suggest that the answer is yes. But this single example, or even an assortment of examples, does not answer the basic question of whether *all* solutions of the differential equation  $y'' = x + y - y^2$  are oscillatory in nature. Also, what is happening to the solution curves in Figure 3.7.2 when

\*Some numerical solvers require only that a second-order differential equation be expressed in normal form  $y'' = f(x, y, y')$ . The translation of the single equation into a system of two equations is then built into the computer program, since the first equation of the system is always  $y' = u$  and the second equation is  $u' = f(x, y, u)$ .



**FIGURE 3.7.1** Comparison of two approximate solutions in Example 4



**FIGURE 3.7.2** Numerical solution curve of IVP in (1) of Example 3

$x$  is near  $-1$ ? What is the behavior of solutions of the differential equation as  $x \rightarrow -\infty$ ? Are solutions bounded as  $x \rightarrow \infty$ ? Questions such as these are not easily answered, in general, for nonlinear second-order differential equations. But certain kinds of second-order equations lend themselves to a systematic qualitative analysis, and these, like their first-order relatives encountered in Section 2.1, are the kind that have no explicit dependence on the independent variable. Second-order ODEs of the form

$$F(y, y', y'') = 0 \quad \text{or} \quad \frac{d^2y}{dx^2} = f(y, y'),$$

that is, equations free of the independent variable  $x$ , are called **autonomous**. The differential equation in Example 2 is autonomous, and because of the presence of the  $x$  term on its right side, the equation in Example 3 is nonautonomous. For an in-depth treatment of the topic of stability of autonomous second-order differential equations and autonomous systems of differential equations, the reader is referred to Chapter 11.

### 3.7 Exercises Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1 and 2, verify that  $y_1$  and  $y_2$  are solutions of the given differential equation but that  $y = c_1y_1 + c_2y_2$  is, in general, not a solution.

1.  $(y'')^2 = y^2$ ;  $y_1 = e^x$ ,  $y_2 = \cos x$
2.  $yy'' = \frac{1}{2}(y')^2$ ;  $y_1 = 1$ ,  $y_2 = x^2$

In Problems 3–8, solve the given differential equation by using the substitution  $u = y'$ .

3.  $y'' + (y')^2 + 1 = 0$
4.  $y'' = 1 + (y')^2$
5.  $x^2y'' + (y')^2 = 0$
6.  $(y + 1)y'' = (y')^2$
7.  $y'' + 2y(y')^3 = 0$
8.  $y^2y'' = y'$
9. Consider the initial-value problem

$$y'' + yy' = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

- (a) Use the DE and a numerical solver to graph the solution curve.
- (b) Find an explicit solution of the IVP. Use a graphing utility to graph this solution.
- (c) Find an interval of definition for the solution in part (b).
10. Find two solutions of the initial-value problem

$$(y'')^2 + (y')^2 = 1, \quad y(\pi/2) = \frac{1}{2}, \quad y'(\pi/2) = \sqrt{3}/2.$$

Use a numerical solver to graph the solution curves.

In Problems 11 and 12, show that the substitution  $u = y'$  leads to a Bernoulli equation. Solve this equation (see Section 2.5).

11.  $xy'' = y' + (y')^3$
12.  $xy'' = y' + x(y')^2$

In Problems 13–16, proceed as in Example 3 and obtain the first six nonzero terms of a Taylor series solution, centered at 0, of the given initial-value problem. Use a numerical solver and a graphing utility to compare the solution curve with the graph of the Taylor polynomial.

13.  $y'' = x + y^2$ ,  $y(0) = 1$ ,  $y'(0) = 1$
14.  $y'' + y^2 = 1$ ,  $y(0) = 2$ ,  $y'(0) = 3$
15.  $y'' = x^2 + y^2 - 2y'$ ,  $y(0) = 1$ ,  $y'(0) = 1$
16.  $y'' = e^y$ ,  $y(0) = 0$ ,  $y'(0) = -1$
17. In calculus, the curvature of a curve that is defined by a function  $y = f(x)$  is defined as

$$\kappa = \frac{y''}{[1 + (y')^2]^{3/2}}.$$

Find  $y = f(x)$  for which  $\kappa = 1$ . [Hint: For simplicity, ignore constants of integration.]

### Discussion Problems

18. In Problem 1 we saw that  $\cos x$  and  $e^x$  were solutions of the nonlinear equation  $(y'')^2 - y^2 = 0$ . Verify that  $\sin x$  and  $e^{-x}$  are also solutions. Without attempting to solve the differential equation, discuss how these explicit solutions can be found by using knowledge about linear equations. Without attempting to verify, discuss why the linear combinations  $y = c_1e^x + c_2e^{-x} + c_3 \cos x + c_4 \sin x$  and  $y = c_2e^{-x} + c_4 \sin x$  are not, in general, solutions, but the two special linear combinations  $y = c_1e^x + c_2e^{-x}$  and  $y = c_3 \cos x + c_4 \sin x$  must satisfy the differential equation.
19. Discuss how the method of reduction of order considered in this section can be applied to the third-order differential equation  $y''' = \sqrt{1 + (y'')^2}$ . Carry out your ideas and solve the equation.
20. Discuss how to find an alternative two-parameter family of solutions for the nonlinear differential equation  $y'' = 2x(y')^2$  in Example 1. [Hint: Suppose that  $-c_1^2$  is used as the constant of integration instead of  $+c_1^2$ .]